

Kenya County Climate Risk Profile: Nandi County

Highlights

- Nandi County is located in the North Rift Region of Kenya (Figure 1). The county experiences moderate dry spells with around 50 mm of rainfall in the months of January and February. April receives the most rainfall, more than 200 mm per month.
- Agriculture is the largest contributor to the economy of Nandi County. About 75% of households are engaged in farming activities, and 54% of the county's labour force is engaged in the agricultural sector.
- Despite the importance of agriculture in Nandi County, farmers continue to depend on rain-fed agricultural systems, which are erratic and unreliable. This situation has contributed to food insecurity in the county. The food poverty rate in Nandi County stands at almost 47%.
- The effects of climate change in Nandi are extensive, affecting approximately 83% of farming households. Excess rainfall, low temperatures, frost and hailstorms, and changes to the growing seasons are the most prevalent climatic hazards in the county.
- Crop and livestock farmers have adapted to the changing climate. Several organizations support farmers through off-farm interventions and services.
- The county's adaptive capacity is hindered by low income, poor road infrastructure, poor market linkages, mismanagement, a lack of education, inadequate policies, and inadequate budgeting.
- A current climate change policy proposal seeks to form a Nandi County climate change department and allocate 2% of the county's budget to it.

Figure 1: Map of Nandi County

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List of Acronyms

AI - Artificial insemination

ASDSP - Agricultural Sector Development Support Program

CDDs - Consecutive dry days

CIDP - County Integrated Development Plan

CMIP5 - Coupled Model Intercomparison Project Phase 5

ha - Hectare

kg - Kilograms

GoK - Government of Kenya

IPCC [AR5] - Intergovernmental Panel on Climate Change Fifth Assessment Report

KALRO - Kenya Agricultural and Livestock Research Organization

KSh - Kenyan shillings

KTDA - Kenya Tea Development Agency

mm - millimeters

NT35 - Number of days with maximum temperatures greater than or equal to 35°C

P5D - Average rainfall over 5 days

Nandi

Foreword

The mandate of the Ministry of Agriculture, Livestock, Fisheries and Co-operatives is to create an enabling environment for sustainable development of agriculture and co-operatives for economic development. This objective underpins our desire and commitment to transform Kenya into a newly industrializing, middle income country providing a high quality of life to all its citizens in a clean and secure environment as envisaged in our development blueprints, the Kenya Vision 2030, the Big Four Agenda and the Agricultural Sector Transformation and Growth Strategy (ASTSG 2019 – 2029). The sector remains high on the national development agenda in terms of food and nutrition security, income generation, employment creation, saving and investment mobilization and export earnings. To realize the country's aspirations of food and nutrition security, the Government through this Ministry is implementing the National Agricultural and Rural Inclusive Growth Project (NARIGP) with the support of the World Bank. The development objective of the project is to increase the agricultural productivity and profitability of targeted rural communities in 21 counties and in the event of an eligible crisis or emergency, provide an immediate and effective response.

The agriculture sector is however, highly vulnerable to the impacts of climate change and extreme weather events. Responses that would enable the country to cope with these risks are outlined in the Kenya Climate-Smart Agriculture (CSA) Strategy and in the commitments of the Kenya Nationally Determined Contributions (NDC) to the United Nations Framework Convention on Climate Change (UNFCCC). In 2010, the Government developed the National Climate Change Response Strategy (NCCRS) which recognized the impacts of climate change on the country's development. This was followed by the development of the National Climate Change Action Plan in 2012. The focus of these initiatives include the development of county-level climate risk profiles to mainstream climate change perspectives in programs and development plans at county level. The Ministry has developed county climate risk profiles in 31 counties and NARIGP is supporting the development of profiles for an additional 14 counties. The purpose of the profiles is to inform county governments and stakeholders on the climate change risks and provide opportunities for integration into respective county development plans and processes.

This climate risk profiles study will be used as a basis to climate proof projects or any other developments in fourteen counties (Samburu, Turkana, Kitui, Narok, Kirinyaga, Kiambu, Muranga, Bungoma, Trans Nzoia, Nandi, Vihiga, Kisii, Nyamira and Migori). The study provides information on current and possible future climate scenarios, climate-related vulnerabilities and risks for key major agricultural value chains, policy landscape and the institutional capacity to deliver adaptation programs. Each profile presents adaptation and risk reduction options that can transform and reorient agricultural systems in the counties to increase productivity, enhance smallholder farmers' resilience and mitigate against climate change.

Finally, I call upon all stakeholders for their cooperation and support for adoption of CSA production practices that maximize the triple wins: increases productivity, enhanced resilience and reduced greenhouse gas (GHG) emissions. Through the adoption of new technologies and improved practices, we will realize the desired goal of Kenya being a food and nutrition secure country, fostering socio-economic development and improved livelihoods of Kenyans.



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1. Introduction

Climate change is becoming one of Kenya's most serious threats. The country is susceptible to climate-related events, and projections indicate that such events will continue to affect Kenya in the future. In many areas, extreme weather is now the norm. Rainfall is irregular and unpredictable; some regions experience frequent droughts during the long rainy season or severe floods during the short rainy season. Arid and semi-arid areas are particularly vulnerable to these extreme changes, putting the lives and socioeconomic activities of millions of households at risk.

The Kenya Vision 2030 is a national blueprint that seeks to transform Kenya into a middle-income country that provides a high quality of life and a clean and secure environment to all its citizens by 2030. This blueprint has identified the agricultural sector as a key contributor to Kenya's economic growth. However, the agricultural sector is constrained by inadequate access to quality goods, marketing inefficiencies, a subpar investment environment, decreasing soil fertility, insufficient mechanization, land fragmentation, and - most significantly - climate change.

In 2010, Kenya developed a National Climate Change Response Strategy which recognized the impact of climate change on the country's development. This was followed by the 2012 National Climate Change Action Plan, which provided a means for implementing response strategies and highlighted the country's priorities. These two initiatives are focused on the national level, but the response to climate change also needs to be integrated into county-level policies, programs, and development plans. Initiatives need to be locally relevant and actively involve local stakeholders.

Through the Ministry of Agriculture, the Government of Kenya (GoK) is implementing the National Agricultural and Rural Inclusive Growth Project (NARIGP) with support from the World Bank. The project's objective

is to increase the agricultural productivity and profitability of targeted rural communities in selected counties. To address the climate change risks and vulnerabilities that negatively impact agricultural production, the Alliance of Bioversity International and the International Center for Tropical Agriculture (CIAT) completed a climate risk assessment in 14 counties supported by NARIGP. The aims of the assessment are to provide information about the current climate and possible future climate scenarios, to pinpoint climate-related vulnerabilities and risks for major agricultural value chains and specific groups of people involved in agriculture, to identify adaptation options that address climate risks and vulnerabilities, and to assess the institutional capacity to deliver adaptation programs.

This climate risk profile seeks to inform county governments and stakeholders about climate change risks and opportunities for agriculture so they can integrate these perspectives into county development. This report will help county governments and stakeholders integrate climate change risks and opportunities for local agriculture into county development plans.

The Alliance undertook the assessment in a set of interrelated stages (Figure 1). It first initiated a desk review of the conceptual and analytical contexts of climate change risks at the national and county levels. It made efforts to involve a wide range of institutions that have worked on climate change at the national and regional levels. The team used globally available data sources like the Kenya Open Data Portal and county development plans, and collected information from relevant government departments, such as the Department of Resource Surveys and Remote Sensing, the Kenya Meteorological Department, and the Drought Monitoring Centre. The team also collected data through focus group discussions, key informant interviews with carefully selected experts, climate modeling, and three days of sub-national stakeholder workshops. The final reports were then presented and validated by national- and county-level stakeholders.

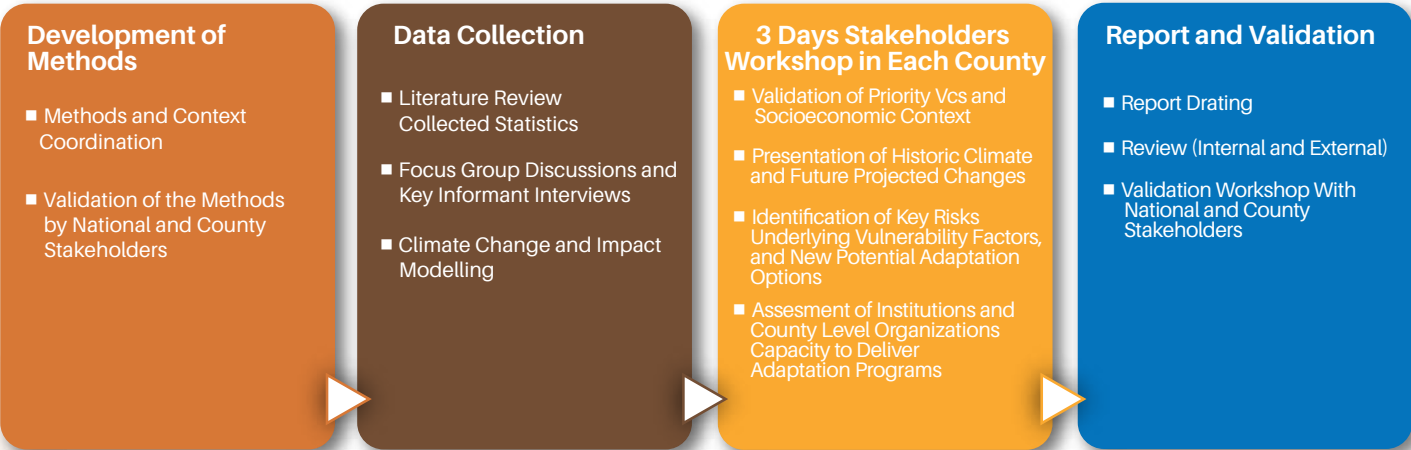


Figure 2: Stages in developing a climate risk profile

2. County Context

The landscape of Nandi County is generally hilly. The county is split into five areas: rolling hills in the west, the Kapsabet Plateau, the Tinderet Volcanic mass in the southeast, the King'wal Swamp in the center, and the Nyando Escarpment at the southern border (County Government of Nandi 2018).

2.1 Economic Relevance of Farming

The agricultural sector is the Nandi County's biggest employer. About 54% of the total labor force aged 15-

2.2 People and Livelihoods

The county has an absolute poverty level of 48% (County Government of Nandi 2013). This accounts for 3% of the national poverty rate. The impoverished population (40%) especially living below the poverty line (US\$ 1.90 a day) is mostly composed of widows, orphans, female headed households, people living with disabilities, and small-scale, illiterate farmers (County Government of Nandi, 2018). These vulnerable groups have limited or no access to land or balanced diets. Currently, 46% of residents in Nandi County have access to potable water from sources like protected springs and wells, boreholes, water pipes, rainwater collection, bottled water, and public taps. The rest (55%) rely on sources like ponds, dams, lakes, streams, rivers, unprotected springs and wells, and water vendors (KNBSb, 2019). Streams and rivers are a major water source, with 37% of households relying on them. The average rural area is 0.5 km away from the nearest water source (County Government of Nandi, 2018). In Nandi County, 31% of the population uses electricity as a main source of lighting, and 29% uses the sun. Currently, 85% of households rely on firewood for cooking, 0.5% use electricity, 6% use liquefied petroleum gas, and 8% use charcoal (KNBSb, 2019). Many are reliant on firewood because Nandi County is home to many forests and lacks clear forest protection policies; in addition, other energy sources are inaccessible.

Livelihoods and agriculture in Nandi

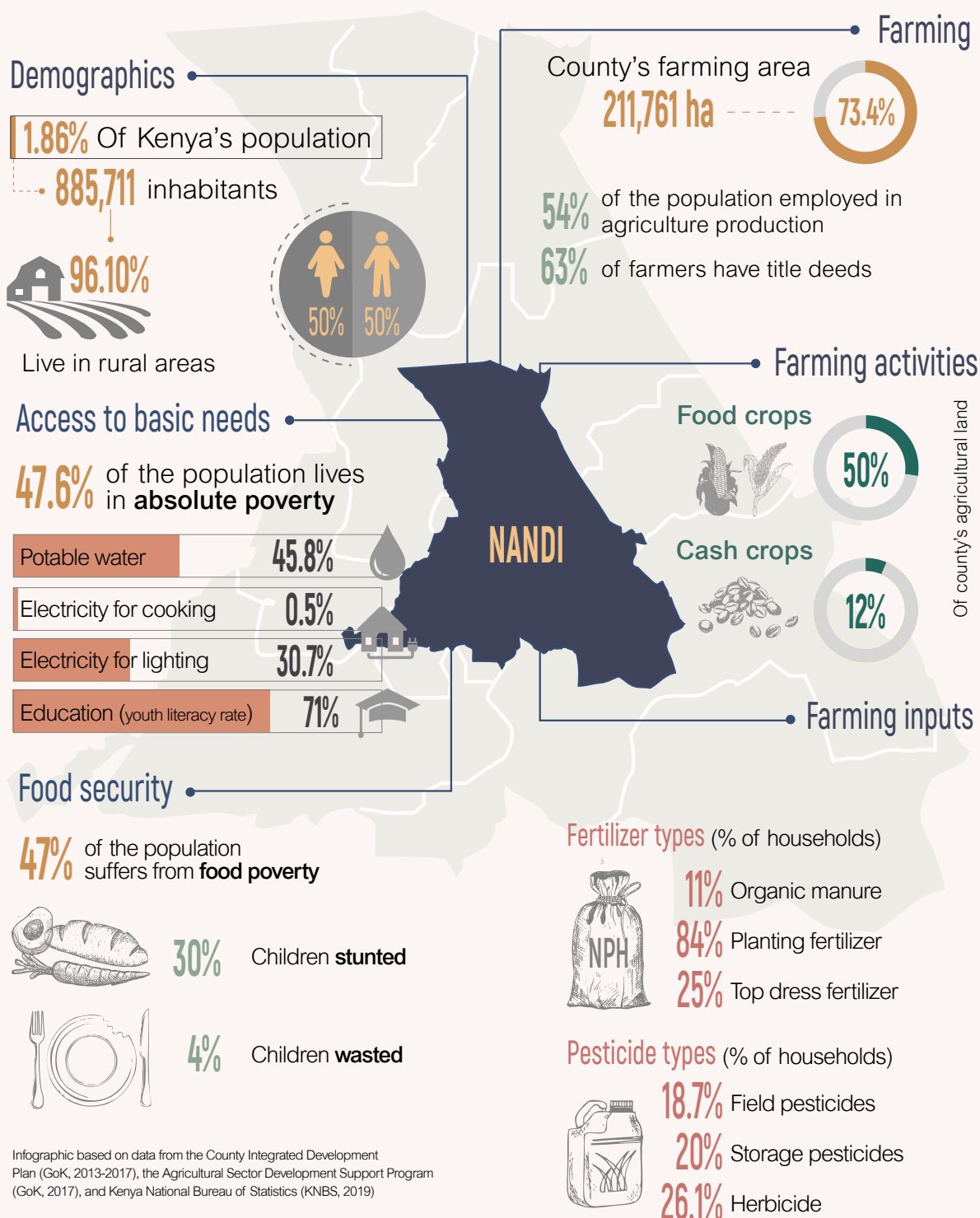


Figure 3: Agriculture and livelihoods in Nandi County

A total of 148,887 households or 75% of the county's population is engaged in farming activities. Out of this population, 96% or 143,017 households are involved in crop farming, and 75% or 111,347 households practice livestock farming. About 1% of farmers practice aquaculture and fishing (KNBSb, 2019).

2.3 Agricultural Activities

Nandi County has a total land area of 288,440 ha, of which 211,761 ha (about 73%) is arable. Its land area accounts for approximately 0.5% of the total land area in Kenya. The county's agriculture is primarily rain-fed. The main food crops are maize, beans, finger millet, sorghum, sweet potatoes, and cassava. Food crops cover a total of 105,087 ha. The primary cash crops are tea, coffee, and sugar cane. Cash crops cover a total of 26,290 ha. The key livestock are cattle and poultry. Farmers also keep goats, sheep, pigs, and bees (County Government of Nandi, 2018).

There are three major agroecological zones in Nandi County: the Upper Highlands, the Lower Highlands, and the Upper Midlands (Figure 3). Generally, the county receives an average annual rainfall of about 1200-2000 mm (County Government of Nandi, 2013). The regions that experience 1500 mm or more per

annum constitute the extended agroecological zones conducive to tea cultivation in the Upper and Lower Highlands. The relatively drier areas to the east and northeast, which receive an average of 1200 mm per annum, are suitable for maize, sugarcane, and coffee. Forests account for 12% of the county's total land area. Nandi County contains six gazetted forest reserves that cover an area of 52,425 ha. The largest is the South Nandi Forest, which is a tropical rainforest that covers an area of 20,150 ha. Another tropical rainforest, the North Nandi Forest, encompasses an additional 16,004 ha.

Nandi County is characterized by small-scale farms. The average small-scale farm is 1.32 ha in size, while the average large-scale farm spans 11.2 ha. Some farmers, especially those who cultivate tea, hold large plots of land measuring 40 - 400 ha or more. Small-scale farms grow both subsistence and commercial crops, while large-scale farms are mainly commercial. The rampant subdivision of agricultural land has fragmented farming plots to as small as 0.1 acre (County Government of Nandi, 2018). In Nandi County, land is either freehold or leasehold. About 63% of farmers hold title deeds for their land (County Government of Nandi, 2013). Freehold tenure applies to customary land, which is usually put to agricultural use. Large farms including tea plantations, cooperative farms, and large-scale farmers usually operate on leased land.

As regards input use in Nandi County, 84% of farming households apply fertilizer, 25% of farming households use top-dressing fertilizer, and 11% of farming households utilize manure. Only 2% of farmers have irrigation systems, while the rest depend on rain-fed agriculture. Approximately 19% of households use field pesticides, 20% employ storage pesticides, and 27% apply herbicides. With perennial crops, 47% of farmers use top-dressing fertilizers, 37% apply basal fertilizers, 31% utilize herbicides, 14% employ organic manure, 3% use foliar feeds, and 3% apply field pesticides (ASDSP, 2014).

2.4 Agricultural Value Chains

Nandi County enjoys a broad diversity of agricultural production systems. Various value chains have been prioritized for development interventions by different government organizations and programs, such as the County Integrated Development Plan (CIDP) and the Agricultural Sector Development Support Program (ASDSP), and government institutions such as the Kenya Agricultural and Livestock Research Organization (KALRO). For the development of this profile, a list of the major agricultural value chain commodities (VCCs) in Nandi County was compiled using the following prioritization indicators: productivity characteristics, including harvested area, production, and production variations in the past five years; economic value (KSh); and nutrition characteristics like dietary energy consumption (Kcal/capita/day) and protein, iron, zinc,

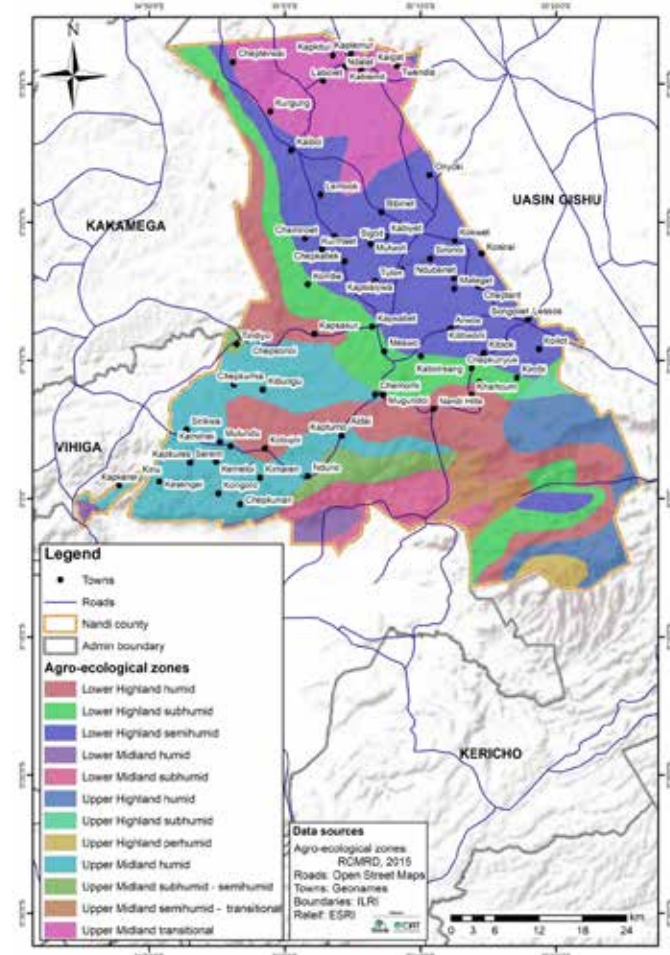


Figure 4: A map of agroecological zones in Nandi County

and vitamin A content. The team presented this list to stakeholders during a three-day workshop for in-depth analysis and selection. We further honed the selection using a set of criteria which were first determined with the stakeholders. The criteria were resilience to current and future climate change impacts, on a scale from low to high; the percentage of population involved in the value chain (%); and involvement in the value chain of economically and socially vulnerable groups such as poor people, women, and youths, again measured on a scale from low to high. First, we assessed each value chain against each criterion. Then, we selected the value chains that involved the highest percentage of the population and engaged poor people, women, and youths. The four VCCs chosen for this report are maize, tea, dairy cattle, and indigenous chickens (Figure 5).

2.4.1 Maize

Each year, Nandi County produces approximately 2 million bags of maize and consumes approximately 900,000 bags (Owuor, 2020). The excess is sold within or outside the county. The total value of maize in Nandi County is approximately KSh 4.75 billion (County Government of Nandi, 2013). About 90% of the farming population, or 134,408 households, are engaged in maize production (KNBSb, 2019). Maize is primarily produced during the long rainy season and is considered a staple food for most households. Farmers who produce maize are mainly small and medium-scale, with an average farm sizes of 1.2 ha. However, some large-scale farmers cultivate maize on plots of more than 4 ha of land.

Large-scale distributors like the Kenya Seed Company and the Western Seed Company supply seeds to agro-dealers, who sell seeds and other inputs to farmers. Agro-dealers mainly operate on a small or medium scale. Approximately 90% of farmers use certified maize seeds, and 84% of farmers use inorganic basal fertilizers in Nandi County (ASDSP, 2014).

Women are the main laborers in on-farm activities like land preparation, planting, weeding, and harvesting. Clearing weeds and bushes for land preparation is done either manually, or with tractors and herbicides. Planting is a critical stage in maize production and normally occurs at the onset of the rainy season. Early planting allows maize seeds to benefit from warm temperatures, good aeration, and fewer pests and diseases. Harvesting is usually done manually, relying on hired labor. About 10% of the crop is harvested for home consumption or sale, primarily due to food shortages in the early season (County Government of Nandi, 2013). Early harvest reduces the final amount of maize brought in at the end of the season.

Men and youths are significantly involved in post-production activities like threshing, drying, and storage. Transportation for storage and marketing is a key activity at this stage but is threatened by poor

road networks. Bulking occurs in homes and in farmers' cooperative organizations. The Kenya Cereals Enhancement Program has created warehouses in Chesumei and Mosop to store maize that is awaiting processing. The Kenya Cereals Enhancement Program also provides farmers with equipment like solar dryers, which increase the shelf life and reduce the vulnerability of maize grains. Farmers are also beginning to store grains in hermetic bags in order to use fewer storage pesticides. Posho Mills are the primary processors of maize flour for home consumption.

Maize is mainly sold in the form of threshed and dried grains. About 50% of the total maize produced in Nandi County is sold to different markets. Maize markets are dominated by middlemen. Small-scale farmers who produce minor quantities and require cash for immediate household needs sell their produce to middlemen. Other buyers include the National Cereals and Produce Board, millers, and institutions. The price of maize is highly volatile due to changes in production across different seasons.

2.4.2 Tea

Tea is the leading export of Kenya. As previously stated, Nandi County is the country's third-largest producer of tea, which is its major cash crop. The county produces 181,000 tons per year, and a large portion (41-60%) of the population in Nandi is involved in tea production. The small-scale production of tea is managed by the Kenya Tea Development Agency (KTDA) and run by autonomous tea factories, while multinational producers are owned and managed as private commercial enterprises under the umbrella of the Kenya Tea Growers Association. Some of the multinational companies in Nandi County are Eastern Produce Kenya, Ltd., Williamson Tea, and Nandi Tea Estate. Tea is largely grown in the Nandi Hills, Emgwen, and Aldai Sub-Counties.

Important activities at the input stage of this value chain include acquiring labor, cuttings, and fertilizers. Tea cuttings come from the KTDA and individual and multinational-owned nurseries. Other inputs, such as phosphate fertilizers, are supplied by small and medium-scale agro-dealers. The KTDA provides fertilizers to registered small-scale farmers at subsidized rates. The Tea Research Institute (TRI) focuses on the development of improved tea clones and appropriate technologies for improving yields. Men are more engaged in the input supply stage than women and youths.

At the on-farm production stage, women participate at the highest rate. Major on-farm activities include preparing land, planting, weeding, pruning, applying fertilizer, and plucking. After transplanting, it takes about two years before the first leaf harvesting. Proper management of young tea cuttings is essential for healthy tea bushes. Proper weeding, application of phosphatic fertilizer for good root formation, and

pruning are among the good agricultural practices necessary for tea field establishment.

After plucking, the tea leaves are sorted to obtain the best-quality leaves. The leaves are then taken to a collection center, where they are weighed, then sold and transported to factories for processing. The main factories in Nandi County, the Chebut Tea Factory and the Kaptumo Factory, are managed by the KTDA. These use firewood fuel for their steam boilers. All tea processors in the county are large-scale. Processing facilities range from private factories to multinational-owned factories, to KTDA-owned factories. Black tea is their major output. The youths of Nandi County are highly involved at the post-production stage, especially in the factories.

The bulk of processed tea is sold at auction in the city of Mombasa. Packaging and branding are important activities at this stage, and youths and men are highly involved. At the output market stage, youths participate intensively, and most revenue collection is done by men, especially in male-headed households, as they predominantly control the household finances. Tea hawking, whereby brokers abuse the system to disadvantage farmers, poses a challenge to this stage of the value chain.

2.4.3 Dairy Cattle

Dairy comprises one of the most important value chains in Nandi County. Nearly every rural home (81-100%) owns a milking cow either for subsistence or commercial purposes. There are approximately 232,000 animals in the county, of which 54% are exotic breeds, and 46% are indigenous breeds (KNBSb, 2019). Roughly 121,466,250 kg of milk is produced each year, valued at KSh 3.6 billion (County Government of Nandi, 2018).

The crucial activities at the input stage of this value chain include the acquisition of feed including fodder, mineral supplements, concentrates, and water; extension services and capacity building; and artificial insemination (AI) services. Small and medium-scale agro-dealers supply feeds, milking equipment, drugs, and other inputs. Agro-dealers source these inputs from large-scale manufacturers like Unga Feeds Company and Cooper K Brands. The Nandi County government offers subsidized AI services. The Department of Agriculture provides extension through extension officers, cooperative societies, and development partners like Heifer International and Digicow. Farmer field schools also play a crucial role capacity building.

According to the findings from the stakeholder workshop, about 60% of dairy farmers in Nandi County are small-scale, 35% are medium-scale, and 5% are large-scale. At the on-farm stage, feeding is the most important activity. A dairy cow needs good fodder, minerals, concentrates, and clean drinking water.

Because fodder is so important, farmers proactively undertake fodder establishment across the county, growing grasses like Napier, Brachiaria, and Boma Rhodes on farms; preparing silage; hay making; and preparing maize stover as feed. These activities ensure constant and sufficient availability of fodder throughout the season. However, a large number of dairy farmers still practice open grazing. Animal health management is an integral part of dairy production. Farmers in Nandi County take their cows to cattle dips or perform home spraying against ticks, while veterinary officers are involved in vaccination and disease treatment. Dairy farmers in Nandi County are normally encouraged to join cooperative societies, since these are the vehicles of government support.

The actors involved in milk value addition in Nandi include local traders who own portable milk shops with cooling abilities. Producer organizations, like the Aldai Dairy Cooperative Society or the Kaplamai Dairy Cooperative, process raw milk into yogurt, fermented milk, and processed fresh milk, among other products. Non-processed milk marketing channels in Nandi County include direct milk sales to consumers by farm households, and co-operative societies, self-help groups, and traders also collect raw milk to sell to processors and consumers. In addition, some farmers sell milk directly to processors like Brookside, Kenya Cooperative Creameries, and smaller private processors.

2.4.4 Indigenous Chickens

Indigenous chickens are at the center of a popular household enterprise in Nandi County, where 81% of the population keeps chickens. Most of these chicken farmers are small-scale producers, with each household holding an average of 10 chickens. Nandi County has 700,000 birds (KNBSb, 2019), of which 93% are indigenous chickens, 5% are laying chickens, and 2% are geese and turkeys (County Government of Nandi, 2018).

Good chicken housing forms the foundation of poultry production. Before chicks are brought in, farmers need quality structures to protect the chicks. Chicks are acquired from farmers who hatch them within and outside the county; from large-scale hatcheries like Kukuchicks, Kenchic, and KALRO; and through grants from government initiatives like NARIGP. The government supports the use of improved indigenous breeds like Kenbro, Kuroiler, Kari, and Rainbow Roosters for their higher hatching rate and other benefits. At the input stage, agro-dealers supply inputs like feeds, vaccines, and antibiotics. Both men and women are highly involved in this stage.

Brooding chicks, an on-farm activity, is vital to the indigenous chicken value chain. Good brooding facilities require adequate warmth, feeders, and drinkers. Chicks aged one day to three weeks are kept in brooders. Feeding is also one of the most important

Egg collection and sorting are vital processes that determine prices. Packaging can reduce damage during transit. Chickens and chicken products are mainly sold to shops, homes, restaurants, and institutions like hospitals. The chicken value chain in Nandi County takes in an estimated KSh 37.8 million from the sale of birds and KSh 202.9 million from the sale of eggs each year (County Government of Nandi 2019). Major challenges to this value chain include poor husbandry practices, high rates of chick mortality, and poor or disorganized marketing of birds and eggs.



2.5 Challenges to the Agricultural Sector

Nandi County's agriculture sector faces several challenges, including a reliance on rain-fed agriculture. Changing and unpredictable rainy seasons make it harder for farmers to plan. When rains are delayed or insufficient, the county experiences water shortages. Only 2% of households practice irrigation, mainly because the required infrastructure is very expensive (KNBSb, 2019).

Few farmers in Nandi County use modern farming technologies and animal husbandry. Current research-extension-farmer linkages are insufficient to facilitate demand-driven innovation and heightened use of improved technologies; these limitations constrain agricultural productivity. Access to extension services is suboptimal in most parts of the county, which prevents most farmers from keeping pace with changing technologies.

Pests and diseases also contribute to farmers' losses, which are exacerbated by a lack of knowledge on treatment and control among farmers. Inadequate storage causes post-harvest losses. For example, in some parts of Nandi, maize has been contaminated by aflatoxins due to poor storage facilities. Most farmers also lack information on the right types of inputs to use and their appropriate application timeframes. Key inputs such as seeds, pesticides, fertilizers, drugs, and vaccines are expensive relative to farmers' incomes, so some farmers cannot afford to use them to their full potential or at all, and this setback affects their yields.

Poor road networks in Nandi County hinder market access. Out of its 1,673 km of road, only 183 km are paved (County Government of Nandi, 2013). The unpaved roads are in bad condition especially during the rainy seasons. Mosop and Tinderet Sub-Counties are most limited by the poor road networks, despite their high agricultural productivity. Farmers in these areas incur high costs while transporting their produce to markets.

3. Climate Change and Agriculture: Risks and Vulnerabilities

In generating this profile, we assessed past trends and future projections of precipitation and temperature, and computed several related hazards from these two variables. These hazards included extreme hydrological events like flash floods, droughts, moisture stress, heat stress, and the start and length of the growing seasons, in order to assess climate change and variability in Nandi County. The growing season was defined as follows: the first, long rainy season is the

100-day wettest period from January to June, while the second, short rainy season is the 100-day wettest period from July to December (KMD, 2020).

We used Representative Concentration Pathway (RCP) 8.5, one of the four greenhouse gas concentration trajectories adopted by the Intergovernmental Panel on Climate Change (IPCC) for its fifth Assessment Report (AR5) in 2014. Future climate projections were generated based on an ensemble of multiple Coupled Model Intercomparison Project (CMIP5) models (Taylor et al., 2012), using RCP 8.5 for two future periods, 2030 and 2050.¹

To assess droughts and dry spells, we focused on the maximum number of consecutive dry days (CDD), defined as days receiving rainfall measuring less than 1mm (precipitation < 1 mm day⁻¹). We determined heat stress by measuring the total number of days with maximum temperatures greater than or equal to 35°C (NT35). Growing days are the days during a season when average temperatures are greater than or equal to 5°C and precipitation exceeds half the potential evapotranspiration. The start of the growing season was determined by the occurrence of 5 consecutive growing days, while the length of the growing period (LGP) was determined as the total number of growing days. For each season, heavy precipitation events were captured with a 5-day running average of rainfall, indicative of floods, and the 95th percentile of daily precipitation, suggestive of extremely high rainfall over a short period of time that can lead to events like flash floods. For each pixel, we calculated the 95th percentile of daily precipitation distribution based on the 100 wettest days per season per year.

To assess the degree of adequacy of rainfall and soil moisture to meet the potential water requirements for agriculture, we focused on drought stress, measured by the number of consecutive days in each season when the ratio of actual to potential evapotranspiration falls below 0.5. This value was calculated for each pixel per season per year by evaluating soil's water capacity and evapotranspiration in order to define the number of days that met the requirements for drought stress.

3.1 Climate Change and Variability: Historic and Future Trends

Nandi County's temperatures have historically ranged from 15-28°C (Figures 5 and 6). The long rainy season runs from February to June, and the short rainy season spans August through December. The two seasons bring similar amounts of precipitation. As previously stated, the county experiences moderate dry spells with around 50 mm of rainfall in January and February.

¹For historical precipitation and temperature trends, we used the Climate Hazards Group InfraRed Precipitation with Station (CHIRPS) and Climate Hazards Group Infrared Temperature with Stations (CHIRTS). For future climate projections we used an ensemble of downscaled Coupled Model Intercomparison Project Phase 5 (CMIP5) (Taylor et al., 2012, Navarro-Racines et al 2020), specifically the MOHC_HADGEM2_ES, CESM1_CAM5, GFDL_CM3, MPI_ESM_LR, and MIROC_MIROC5 models

April receives the most rainfall, more than 200 mm per month. On average, Nandi County receives between 1200 and 2200 mm of precipitation annually. Its south western region receives an average of more than 2000 mm of precipitation annually. The northern region is significantly drier and colder than the southern region.

The mean total rainfall remained quite stable in the past (195-2015) for the long rainy season, the projections trends show a similar pattern in the future, with a small decrease for the period 2020-2040 and a small increase for the period 2041-2060 (Figure 8). The trends for the short rainy season show an increase of rainfall for the future. The mean annual temperature trends showed for both season an increase in the past which will continue the future. The long rainy season will become significantly hotter than the short rainy season (Figure 9).

Historically, the growing season has started in February and March. However, future climate projections suggest that the first growing season will begin a month earlier in the eastern part of the county and start up to 10 days later in some western areas. The second growing season is likely to start 2 to 3 months later in some areas.

The LGP in the long rainy season has historically lasted 5 months or less. Future climate projections indicate that the long rainy season LGP will shorten by more than two months throughout the Nandi County. The second growing season is expected to shorten by 1-3 months.

During the long rainy season, most parts of Nandi County historically experienced 15 CDDs. In the future, the county is projected to experience up to 5 more CDDs, which does not suggest a significantly different incidence of drought. During the short rainy season,

most of the county experiences 18 CDDs. In future, the county is projected to experience up to 4 fewer CDDs, suggesting a lower incidence of drought.

High moisture stress negatively affects crop growth. A comparison of historical and future trends indicates that moisture stress is expected to increase across Nandi County from 5 to 15 days in the future (Figure 11). A uniform distribution of moisture stress is expected. During the long rainy season, moisture stress is projected to diminish by up to 20 days in all parts of the county.

The NT35 is an indicator of heat stress. During the long rainy season, the NT35 has historically remained low in Nandi County. Future climate projections indicate the NT35 value will marginally increase in the county's southern regions. The same pattern will be experienced in the second season.

The P5D is an indicator of flood risk. During the long rainy season, the P5D has historically remained below 25 mm in Nandi County. Future climate projections indicate that the P5D will increase, principally in the south, by around 12 mm. This growth in the PD5 suggests a minor increase in flood risk. During the short rainy season, the P5D is expected to increase by 17 mm in the south.

The 95th percentile of daily precipitation value is an indicator of heavy rainfall and is linked with erosion. During the long rainy season, Nandi County's 95th percentile value is usually higher in the south. Future climate projections indicate that the 95th percentile value will remain stable. During the short rainy season, the 95th percentile value will increase significantly (> 10mm), especially in the south and south west where an increase of more than 15mm is projected (Figure 10).

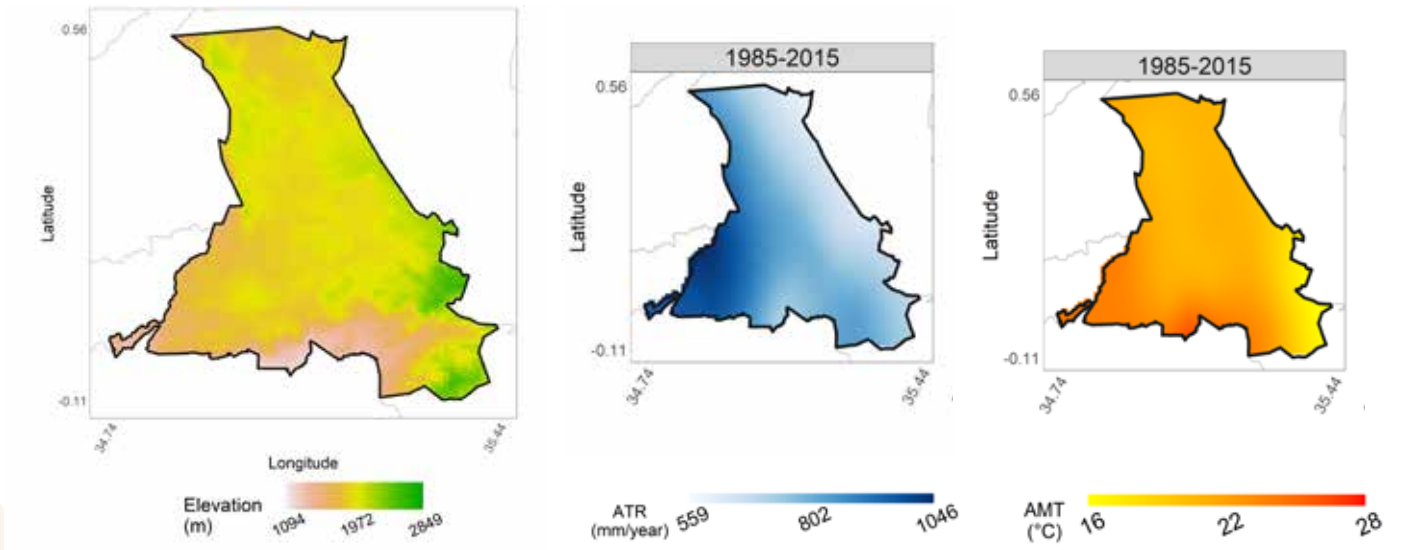


Figure 6: Elevation, average annual precipitation in mm, and average annual temperature in °C in Nandi County (average 1985-2015) for the long rainy season

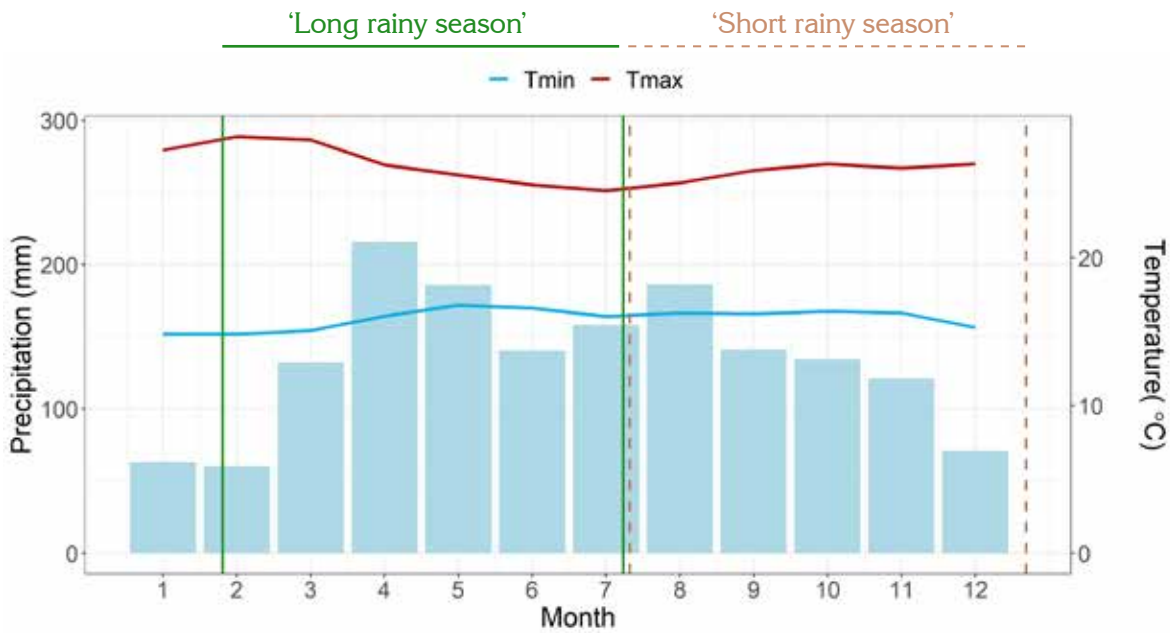


Figure 7: Average monthly temperature and precipitation in Nandi County. The first long rainy season is the 100-day wettest period from January to June, while the second, the short rainy season is the 100-day wettest period from July to December. Bars represent total monthly precipitation and lines represent maximum (red) and minimum (blue) average monthly temperatures (average 1985-2015).

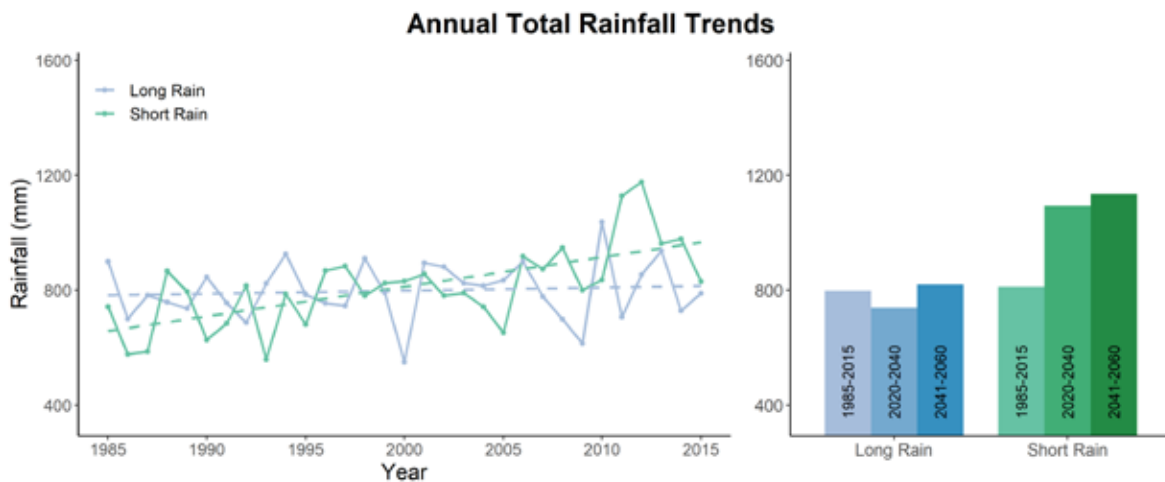


Figure 8: Annual total rainfall trends for the long rainy and short rainy seasons in the past (1985-2015) and in the future (2020-2040 and 2041-2060)

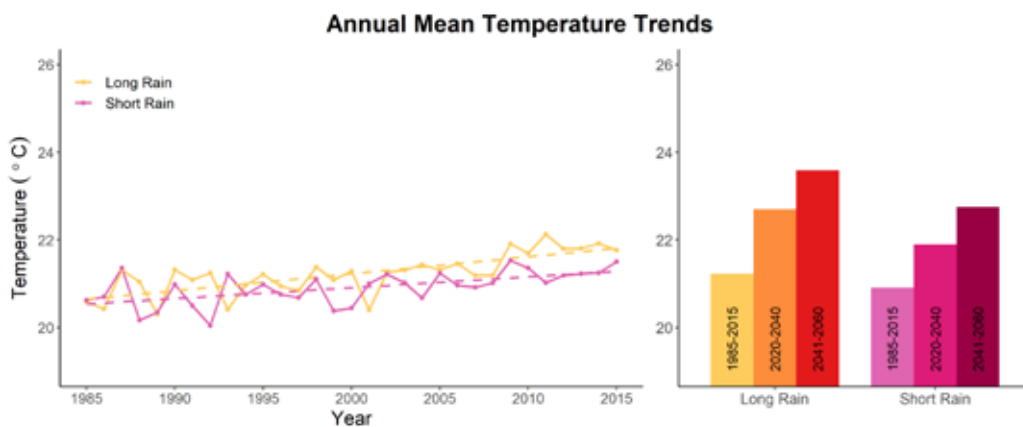


Figure 9: Annual mean temperature trends for the long rainy and short rainy seasons in the past (1985-2015) and in the future (2020-2040 and 2041-2060)

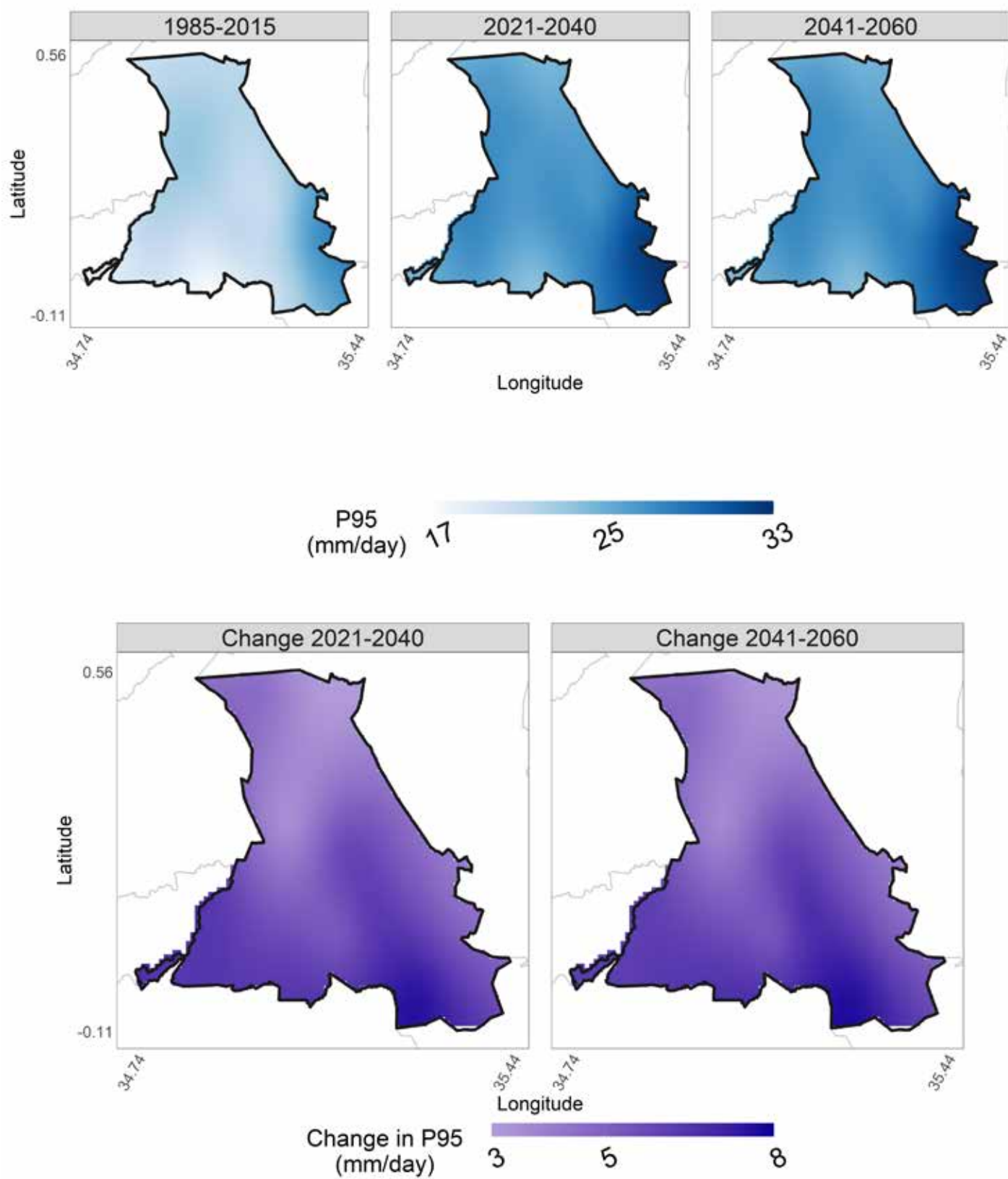


Figure 10: Historical (average 1985-2015), future projected, and projected Change in the 95th percentile of precipitation, an indicator of heavy rainfalls during the short rainy season

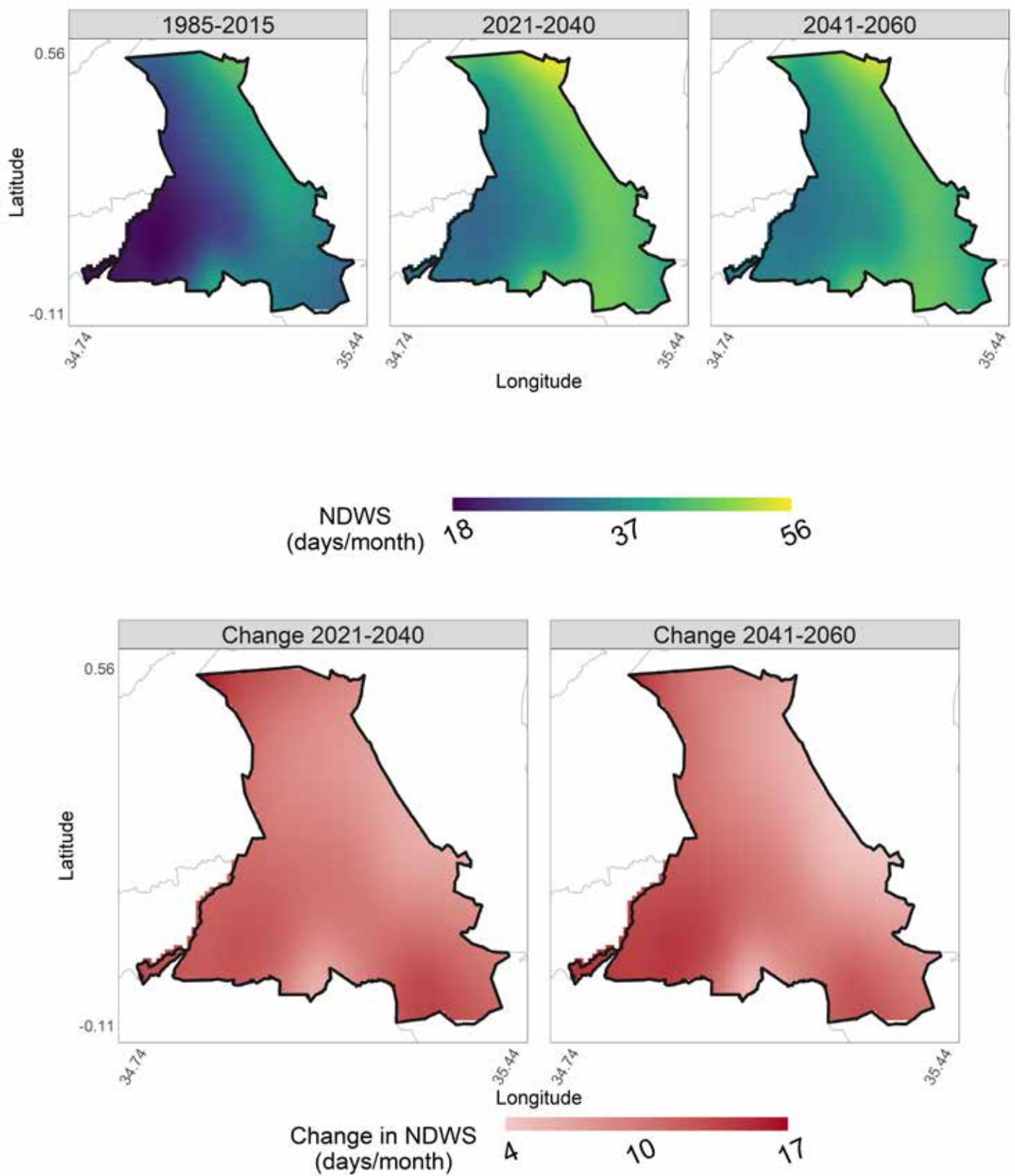


Figure 11: Historical (average 1985-2015), future projected, and projected change in the number of moisture stress days during the long rainy season

3.2 Climate Change from Farmers' Perspectives

Approximately 83% of the farmers in Nandi County have been affected by climate change (ASDSP, 2014). They report that climate change results in altered rainfall patterns and planting seasons, floods caused by excess rainfall, extreme temperatures, hailstorms, factory emissions, soil erosion, the emergence of pests and diseases, the destruction of forests, and landslides, which mainly occur in the Tinderet sub-county. Nandi County farmers receive climate change information via the radio and television, and rely on their knowledge of historical weather patterns. They also receive information from local administration forums commonly referred as “Chief Barazas”, as well as from farmer field school meetings, county departments, social media, and institutions such as the Kenya Agricultural and Livestock Research Organization and the Kenya Tea Development Agency.

Tea farmers in the Emgwen and Nandi Hills areas of Nandi County have suffered hailstorms and frost, which destroy tea leaves on the mother bushes. Farmers from a farmer field school in Emgwen testified that hailstorms destroyed their tea leaves completely; it was up to 60 days before they could harvest again. Frost, on the other hand, leads to leaf drying, especially during dry seasons. These climate hazards cause major losses in both the short and long term. Incidences of excess rainfall, meanwhile, have led to poor utilization of fertilizer due to leaching.

New pests and diseases have proliferated in Nandi County, including fall armyworms that have attacked maize crops in recent years. Pests like fall armyworms are adapted to survive and reproduce rapidly at high temperatures. Thus, more generations are completed during warm spells, which coincide with the crop-growing seasons. Harvesting patterns in the coffee sector have also changed due to variations in weather patterns.

Incidences of extreme temperatures have been harmful to the livestock sector. A young poultry farmer from Nandi Youth Bunge said that he had lost more than half of his indigenous chicken flock within a span of 4 months, due to extremely cold conditions and excess rains. He added that the costs of vaccines and feeds were high during these challenging times, as feed intake and disease incidence are higher in cold seasons. Prolonged dry seasons have reduced the fodder availability for cows, especially for farmers who rely on open-grazing systems and have no conserved fodder.

According to farmer testimonials, most households in Nandi County depend on agriculture for employment. Climate change has contributed to job loss and a reduction of income and wages, especially on tea farms. Farmers living in Tinderet sub-county have

experienced property loss due to landslides and mudslides. Malnutrition is rising among vulnerable groups from food shortages due to these hazards.

3.3 Climate Vulnerabilities across Agricultural Value Chains

With approximately 81% of farming households experiencing effects, climate change poses a grave threat to the agricultural sector of Nandi County (ASDSP, 2014). The main climate hazards that threaten the county include variation in growing seasons, moisture stress, excess rainfall, very low temperatures, and hailstorms.

3.3.1 Maize

The main hazards that affect the maize value chain in Nandi County are moisture stress and excess rainfall. Increased moisture stress reduces vegetative crop growth. Rain delays cause farmers to postpone buying inputs and planting. Maize production is particularly vulnerable, with yield reductions when plants lack the required moisture content levels for effective growth. At the input stage, farmers hold off on the acquisition of inputs like seed, fertilizers, pesticides, and herbicides until the onset of the rains. Delayed planting increases labor costs across all stages of the value chain since most farmers plant at the same time, causing demand for labor to spike. During dry spells, storage pests like weevils proliferate. As a result of moisture stress, farmers have less product to sell. This situation contributes to food insecurity and heightens poverty levels in Nandi County.

Excess rainfall destroys road networks, which makes input acquisition and market access difficult. Continuous heavy downpours delay on-farm operations like land preparation and planting, reducing the growing window. Heavy rains also cause fertilizer leaching and rotting. Storing insufficiently dried maize during excess rains increases the risk of aflatoxin contamination, affecting consumer health and increasing product loss. The correct moisture content for dry maize is 13% (KCEP, 2016). Often maize harvested during heavy rains is of poor quality and attracts low prices from buyers.

3.3.2 Tea

The main hazards that affect the tea value chain are hailstorms, very low temperatures, and frost. Tea is a highly resilient crop. However, climate change interferes with its productivity, and recovery can be a long process. Hail and frost strip leaves off the shrubs, leaving very little for harvest and leading to job losses. Tea seedlings are obtained from mother bushes. If these mother bushes are damaged by hailstorms or frost, it can be difficult to find planting materials for nurseries. Frost also causes leaves to dry up on the bush. Because hailstorms are associated with heavy rains, they lead

to fertilizer leaching, which can increase input costs. Frost and hailstorms lead to high planting costs, as it becomes necessary to repeat farm operations like filling the gaps between damaged seedlings. Hail-damaged canopies can result in increased weed growth, which raises weeding costs and increases fertilizer use in gap-filling. Both hailstorms and frost destroy tea leaves, reducing yield. At the post-harvest stage of the value chain, diminished leaf quality extends sorting time. The costs of transport to factories, marketing, packaging, and branding become elevated because of the low volumes of tea.

The effectiveness and speed with which tea bushes recover from heavy hail and frost damage can affect future crops for several years (Willson, 1992). A major frost event impacted most of the tea-producing counties in Kenya, including Nandi, in 2012, destroying 20 million kg of green leaf and causing an estimated 30% loss in yields (Gachenge, 2012; Bore, 2015).

3.3.3 Dairy Cattle

The major climate hazards that affect dairy production in Nandi County are changing seasons and moisture stress. Unpredictable seasons make for variable fodder production. Extension officers cannot plan for efficient advisories due to alterations in the crop calendar. The calving rate depends on proper feeding. Therefore, unpredictable seasons cause prolonged calving intervals due to inadequate feeds. Moreover, a poorly fed cow is susceptible to parasites and diseases. Unpredictable seasons result in an overlap in the harvest seasons for animal fodder and food crops. Fluctuating feed volumes increase the costs of feed processing and bulking. Inadequately fed cows give less milk, leading to fluctuating milk prices due to changing supply and demand.

Moisture stress limits forage growth, which in turn can mean inadequate feeds. Insufficient feed means low forage yields, reduced calving rates, and depressed milk production. During periods of moisture stress, pests like ticks become more common, resulting in tick-borne diseases like East Coast Fever, and elevating treatment costs. Moisture stress diminishes milk volumes, which lowers revenue. During these times, milk is sold informally, which destabilizes markets. These hazards endanger the viability of the dairy sector, but different stakeholders have intervened to help farmers cope with and adapt to risks.

3.3.4 Indigenous Chickens

Excess rainfall and very low temperatures are the main hazards that affect the indigenous chicken value chain in Nandi County. During excess rains, roads become impassable, which increases transportation costs and the prices of inputs like feeds, antibiotics, and vaccines. Chicks must be rigorously managed during heavy rains, which drive up production costs. Wet days

heighten the incidents of pests and diseases, requiring additional investment in antibiotics and vaccines. Cold and wet weather raises feeding and management costs. Egg quantity and quality are reduced during wet periods, resulting in low prices at the markets. Poultry farmers try to cope by buying high-quality packaging materials during the rainy seasons, which increases their post-production costs. Because markets are often inaccessible during the rainy seasons, middlemen take advantage of farmers and buy eggs at lower prices, reducing farmers' profit margins.

Low temperatures cause high mortality rates among chickens. Most farmers practice a free-range farming system that can leave chickens exposed, and massive losses can occur because most chickens cannot withstand low temperatures. Farmers lack the equipment to provide warmth. Chickens eat more during cold conditions, increasing the cost of feeds. Brooding requires heat, and cold temperatures can raise brooding costs. Incidences of chronic respiratory diseases rise during cold weather, necessitating the purchase of more antibiotics and vaccines. Chilly weather reduces egg production and leads to smaller eggs that fetch lower prices at market.

Vulnerable communities are most at risk from these climate hazards. These individuals are often members of large households with a high demand for food. For example, people with little education do not have information on best practices; they are among the first to experience climate change effects. Farmers without adequate crop diversification suffer when their value chain encounters a hazard. Most rural families have low income levels and small land holdings, which lessens their ability to invest in modern farming technologies. Politically marginalized areas receive less support, rendering them susceptible to climate risks.

4. Adapting to Climate Change

4.1 Factors that Determine Future Vulnerabilities to and Impacts of Climate Change

Vulnerability to climate change is determined by adaptability. Several economic factors are relevant to the food crops sector, which includes maize farmers, such as inadequate monetary resources that lessen the capacity of farmers to acquire inputs like certified seed, fertilizers, and pesticides. Low incomes have driven up poverty levels and vulnerability to climate change. Likewise, poor roads affect both the input acquisition and the output market stages. Most small-scale farmers are found in rural Nandi where suboptimal roads make accessibility a challenge. Steep terrain, especially in Tinderet sub-county, heightens the risk of mudslides and landslides which can destroy crops. Institutional factors like inappropriate and inadequate information about weather patterns contributes to the susceptibility of different groups. Small-scale farmers

of both genders and all ages are often worst affected by climatic hazards, the first to suffer from food shortages and price volatility.

In tea farming, the effects of climatic hazards are exacerbated by several factors: poor roads that hinder the use of smaller vehicles to collect leaves when tea volumes are low; a lack of knowledge about weather patterns and tea management practices; insufficient diversification to other income-generating value chains; poverty among laborers on tea farms the majority of whom come from vulnerable communities; and inadequate policy to guide local consumption and pricing of tea. As a result, farmers and the tea value chain suffer.

The dairy value chain is affected by economic issues such as inadequate resources, a lack of insurance services, high interest rates, and insufficient capital. The value chain also faces institutional hurdles, such as inadequate subsidies to reduce farmers' costs of production. Weak policies have allowed unscrupulous traders to dictate market prices. Small-scale farmers lack the necessary infrastructure to conserve feeds. Bad roads contribute to in-transit spoilage. Women and youths have less access to resources and are therefore more vulnerable to climate change, especially in male-headed households because men are the primary actors in the output market. Farmers, especially small-scale farmers, are the primary victims of climate change in rural Nandi County. Others include producer organizations, processors, milk vendors, and agro-dealers.

The indigenous chicken value chain is mainly made up of small-scale farms that lack the resources and assets to commercialize their enterprise and cope with climate change. In Nandi County, this value chain receives suboptimal support, because not many grants or subsidies exist in the sector. Capacity building in the indigenous chicken sector is also subpar. Most farmers lack relevant knowledge and skills in brooding, vaccination, and disease treatment. There are not enough frontline extension workers in the county. Most vaccines are sourced from distant locations like the town of Eldoret, which means that they are expensive to transport. Poor road networks, in addition, hamper the transportation of feeds and chicks to farms, and of eggs and chickens to markets. Furthermore, there are no policies to set prices for chicken products in Nandi County, which sometimes forces farmers to sell to middlemen at low prices. There are very few producers' organizations in this value chain; this dearth erodes the bargaining power of input suppliers and farmers. Women and youths are the main participants in poultry farming. Their limited access to resources renders them disproportionately vulnerable to climate hazards. Agro-dealers, transporters, and buyers are also affected.

4.2 Adaptation Options

Farmers in Nandi County have adopted certain measures to increase their resilience to climate change. The processes of adapting to the negative effects of climate change and harnessing its potential positive effects is specific to local contexts and circumstances (Figure 11).

4.2.1 Ongoing Adaptation Practices

Farmers have embraced tree planting as an adaptation strategy. About 56% of households in Nandi County plant trees to alleviate the effects of climate change (ASDSP, 2014). The county is removing eucalyptus trees from wetlands and replacing them with indigenous trees. In partnership with the Kenya Forest Service and key stakeholders, the county government aims to establish forest cover of over 10% of the county's area through afforestation programs. Agroforestry is practised to increase the organic matter and available nutrients in the soil and thus increase productivity. The *Grevillea robusta* tree species is preferred for planting in farmlands. *Grevillea* trees are also a source of firewood and timber and can be used as an alternative source of energy (County Government of Nandi, 2018).

The popularity of conservation agriculture is increasing in Nandi County. About 46% of households practice soil and water conservation strategies (ASDSP, 2014). Conservation agriculture practices include using minimum tillage, permanent soil cover, and crop rotation strategies. More than 90% of households have some knowledge about conservation agriculture practices, including ridge cultivation, planting in rows, planting hedgerows, applying manure, crop rotation, and timely weeding (Wambugu et al., 2014). Nearly all farmers apply fertilizer in their fields. Farmers also diversify their crop species and varieties to protect against crop failure. Mixed farming and crop diversification are common adaptive practices in Nandi County (Wambugu et al., 2014). Farmers gravitate towards practices with low capital and labor requirements. Crop rotation and the use of cover crops are the most common.

Irrigation is an important technology practised by a total of 3,777 households in Nandi County, or 2% (KNBSb, 2019). Only 250 ha of land is irrigated. The rest of the agricultural sector depends on rainfall, because irrigation systems are expensive, which contributes to their low uptake. The county government advocates for and supports initiatives geared towards expanding irrigation. Strategies such as exchanging crops for short-maturing types and cultivating more resilient crops like sorghum and millet are used by 42% of households (ASDSP, 2014). Farmers are moving from traditional maize farming to crops like horticulture and potatoes, which have a short growing period and fetch better prices.

automated temperature regulators, and solar heaters.

A key off-farm adaptation strategy in Nandi County is the establishment of farmer cooperatives. These producer organizations increase bargaining power at the input and marketing stages. The county government encourages all farmers to join cooperatives. Currently, there are at least 177 agro-based cooperative societies in the county, including cooperatives geared to dairy, coffee marketing, horticulture, sugar cane marketing, fish, cereal marketing, and tea marketing (County Government of Nandi, 2018). The cooperative movement is weakened by an insufficient regulatory framework, the high cost of credit facilities, inadequate education, rudimentary services, poor leadership and governance, and inadequate extension services. These constraints serve as disincentives to membership.

4.2.2 Potential Adaptation Practices

To protect farmers from climate-related risks, the county could support insurance schemes and educate farmers on the importance of insuring their crops and livestock. The Nandi County Government has partnered with Acre Africa to insure farmers against crop damage caused by unfavourable weather conditions (County Government of Nandi, 2020). This initiative provides weather-index insurance products for each crop phase. Unlike traditional crop insurance schemes, the insurance company does not visit a farmer's field to determine premiums or assess damages; if the rainfall or any other agreed-upon index exceeds a pre-specified range, then the insurance pays out automatically. The uptake of insurance services is, however, still quite low, a situation attributable to limited knowledge about insurance products and depressed incomes among smallholders. Capacity building and upscaling are necessary in the arena of insurance products. Producer organizations can play a critical role in educating farmers about the importance of insuring their farm products.

Subsidy programs are essential in order to boost climate management and adaptation efforts in Nandi County. Subsidies reduce the costs of production and improve livelihoods in the long term. Through the Department of Agriculture, Livestock, Fisheries, and Cooperatives, the county government supports dairy cooperatives with subsidized milk coolers that boost shelf life. Crop farmers receive subsidized inputs like fertilizers through the National Cereals and Produce Board. Other development partners, like the Anglican Development Services and the Kenya Forestry Service, provide free or subsidized fruit and indigenous tree seedlings to farming communities; trees are important carbon sinks. Through programs such as the National Agricultural Rural Inclusive Growth Project, community groups receive grants for projects like growing avocado trees and building water pans for rainwater collection. However, the county still needs more partners to subsidise production factors aimed at the management of and adaptation to climate change.

Extension services provide advisories on farming practices that boost resilience to climate change across agribusiness value chains. Extension teams offer trainings and demonstrations about sustainable land management practices such as conservation agriculture, intercropping, terracing, water harvesting, composting, agroforestry, and agroecological crop selection. Farmers are encouraged to organize themselves into common interest groups based on crop and livestock farming systems. The Nandi County government builds capacity with line departments and development partners. They commonly disseminate information through farmer field schools. Extant technical knowledge among farmers could be further integrated with extension services and information from research organizations. The Kenya Meteorological Department provides weather alerts to farmers and other stakeholders along with information about when to prepare for planting seasons and for climate hazards. Due to these initiatives, capacity building has strong potential to improve farmers' adaptability.

Nandi County has nineteen tea factories, all of which rely on wood fuel to run the boilers and other machineries. Alternative fuel is needed in Nandi County's factories. Over reliance on firewood for cooking energy among households also contributes to deforestation. It is therefore necessary to use clean cooking energy, such as energy-saving stoves and liquefied petroleum gas. Nandi County has aligned its goals to the national target of 10% forest cover. To realise this objective, all tree-felling activities need to be managed by strong policies and enforcement.

In the chicken value chain, the Nandi County government could support community interest groups and poultry cooperatives in establishing commercial hatcheries and sourcing chicks locally. Rather than the conventional method of hens sitting on eggs, more farmers could use incubators to increase hatchability rates. Similarly, in the dairy value chain, the county could support and scale activities such as silage making, biogas production, and marketing through producer organisations.

Adaptation strategies used in selected value chains in Nandi County

Tea				
	Provision of Inputs	On-Farm Production	Harvesting Storage and Processing	Product Marketing
 Hail Storm Consequences	Loss of jobs harvesting tea; reduced availability of mother plants; fertilizer costs increased from leaching and reapplication	High planting costs due to repeating farm operations; attendant rainfall leaches fertilizers and increases weeding difficulty; destruction of tea leaves reduces harvest	Damaged leaves increase sorting difficulty; increased transport and processing costs due to reduced tea volumes	Low volumes increase the costs of packing and branding, transportation, and auctioning
Magnitude of Impact	Minor-Severe	Moderate-Major	Major	Minor-Moderate
Farmers' Current Coping Strategies	Off-farm hawking; purchasing tea plant cuttings from non-affected areas; sourcing fertilizer through tea processors	Establishment of on-farm nurseries; chemical weed control; diversification of labor activities among tea pickers; pruning bushes	Selling low-quality tea to middlemen or brokers; laying off laborers	Increasing prices of green tea; transferring the costs of tea bulking
Potential Adaption Options	Saving income to improve on off-farm business; construction of shade nets; government subsidies for fertilizers; formulation of slow-release fertilizers	Crop insurance; use of integrated weed management control; mechanized fertilizer application; crop diversification; income source diversification among pluckers	Crop insurance; pruning; applying fertilizer; making all-weather roads to accommodate smaller vehicles; adopting small, efficient processing equipment	Promoting local consumption through "Build Kenya, Buy Kenya;" pooling resources for transportation costs; seeking alternative markets
Underlying Factors	Economic: most laborers are poor, without other sources of income	Economic: most tea farmers are affected by hailstorms due to insufficient diversification; institutional: most tea farmers lack knowledge on integrated weed management	Economic: lack of investment in smaller tea processing machines to process low volumes; infrastructure: poor roads hinder use of smaller vehicles for collection when tea volumes are low	Policy: lack of policy framework to guide local tea consumption
 Low Temperatures Consequences	Destruction of mother trees leads to job losses; increased costs of tea seedlings; destruction of seedlings in the nursery; reduced demand for fertilizer	High operation costs due repeated planting; increased weed growth; increased fertilizer use; loss of plucking jobs; reduced yields due to frost	High transportation cost due to low volume	Low tea volume increases the costs of packing and branding, transport, and auction
Magnitude of Impact	Moderate-Severe	Moderate-Major	Moderate	Minor-Moderate
Farmers' Current Coping Strategies	Off-farm hawking; insurance; sourcing tea cuttings from non-affected areas; diversification among agro-dealers	Establishment of on-farm tea nurseries; chemical weed control; income source diversification among pluckers	Farm boundaries increase sorting costs; selling to middlemen; laying off laborers	Reducing personnel; reducing the number of transporters; reducing bonuses; reducing staff through pending leaves
Potential Adaption Options	Saving income and improving off-farm businesses; moving nurseries to areas not prone to frost; diversification to other farm products	Crop insurance; use integrated weed management control; planting indigenous trees along farm boundaries	Improved road infrastructure for quick transportation and to accommodate smaller vehicles; redeployment of staff in the factory; timing maintenance of machines with leaves and off-days	Redeployment of staff; bulking, redeployed transport; expanding online marketing
Underlying Factors	Economic: lack of income source diversification; infrastructure: lack of structures to protect tea seedlings e.g., shade nets	Economic: due to lack of income diversification most of the tea pluckers are VMGs; institutional: lack of integrated weed management knowledge	Economic: lack of small-scale processing machines for low volumes of tea; infrastructure: poor roads do not support transport of low volume of tea using smaller vehicles	Policy: lack of policy framework to guide local tea consumption

Maize



Provision of
Inputs



On-Farm
Production



Harvesting
Storage and
Processing



Product
Marketing



Excess Rainfall
Consequences

Poor road networks lead to delays in seed acquisition due to poor road networks; spoilage of seeds during transportation; delayed fertilizer acquisition and spoilage during transportation; delayed acquisition of pesticides and herbicides

Delays in planning and land preparation; delays in harvesting; increased rotting

Transportation delays due to affected road network; increased transportation costs; extended drying period can lead to delayed bulking; rotting due to increased moisture; decreased quality

Low prices due to poor quality; difficulty linking farmers to buyers; inaccessibility of storage facilities; selling delays due to decreased market access and reduced number of buyers

Magnitude of
Impact

Moderate

Major

Major

Moderate-Major

Farmers' Current
Coping Strategies

Cooperatives acquire seeds and fertilizers for farmers in bulk; integrated pest management technology

Zero tillage or minimum tillage; hand planting; delayed planting; use of early maturing varieties; delayed harvesting; use of ventilated stores; stoking maize in the field

Use of animal and human labor; use of collection centers and warehouses, granaries, and smoking

Sorting and grading; selling produce at any available market price; light vehicle transport; selling in bits to hawkers; use of donkeys

Potential
Adaption Options

Increasing the number of farmers in cooperatives; promoting the use of compost/organic manure for small-scale farmers; soil testing and analysis; upscaling conservation agriculture, and integrated pest management

Promotion of conservation agriculture; early land preparation; subsidizing operation costs; providing weather forecast information to farmers; use of early-maturing varieties; coordination between farmers' cooperatives and drivers

Regular maintenance of the road network; timely harvesting at physiological maturity; provision of driers and warehouses through farmers' cooperatives; use of solar driers at farm level; use of modern ventilated stores

Use of tolerant maize varieties; value addition (milling and packaging); aggregation by cooperatives; improved sorting and grading; regular maintenance of access roads; selling through cooperatives; linking cooperatives to financial institutions

Underlying Factors

Infrastructure: poor road networks; **economic:** some farmers are poorer than others; **biophysical:** some farmers live in lower terrain and find it harder to access inputs

Economic: poor farmers are affected more; cultural: farmers like to do what they are used to; **infrastructure:** farmers living in swampy areas are more affected

Biophysical: some farmers, mainly in Tindiret, live in poor terrain; **economic:** some farmers lack funds for machinery and transportation; **infrastructure:** some farmers are affected by poor road networks

Infrastructure: poor road network prevents produce from reaching all markets; **economic:** some farmers are poor and cannot reach all available markets



Moisture Stress
Consequences

Delay in acquisition of certified seeds, fertilizer, pesticides, and herbicides

Insufficient labor supply increases the cost of land preparation; high cost of maintaining farm machinery; delayed planting; poor seed germination; increased harvest costs due to high demand for labor

Transportation costs increases due to spikes in demand; bulking is reduced because of high demand; increased infestation of storage pest due to high temperatures

Low prices; low linkage between farmers and markets; need drives farmers to sell independently; difficulty in selling due to oversupply at produce in the market

Magnitude of
Impact

Moderate

Moderate-Major

Minor-Moderate

Moderate

Farmers' Current
Coping Strategies

Delay acquisition of inputs; selecting seed from farm stores; acquiring fertilizers, insecticides, and herbicides through cooperatives; planting without fertilizers; use of indigenous technical knowledge

Minimizing operations; zero tillage; reducing area under maize; planting in small portions; dry planting; delayed planting; use of family labor; harvesting incrementally

Use of motorbikes, donkeys, and bicycles; aggregating incrementally selling at prevailing price; use of indigenous technical knowledge; use of chemicals (e.g., pesticide dusting); hematic bags for storage

Accepting prevailing price to get money for essential needs; traders buy produce from individual farms; farmers sell incrementally

Potential
Adaption Options

Providing weather information to farmers on a daily basis; providing seeds through cooperatives; promoting the use of organic fertilizers; encouraging farmers to join cooperatives; promoting the use of indigenous technical knowledge among farmers; encouraging conservation agriculture

Promotion of conservation agriculture; facilitating cooperatives to facilitate services for farmers at subsidize rates; Irrigation; planting without fertilizer to minimize seed scorching; providing weather forecast information to farmers; early harvesting; encouraging the use of family labor; encouraging mechanized labor

Promotion of farmers' cooperative membership for transportation, and bulking early in the season to accumulate volumes; upscaling use of hematic bags; metallic silos; improved, well-ventilated storage facilities

Establishing revolving funds in cooperatives to pay farmers amount of produce delivered; warehouse receipt system; encouraging farmers to sell through cooperatives

Underlying Factors

Economic: some farmers lack capital to purchase inputs on time; **institutional:** farmers do not access information on weather patterns to inform them about the start of rains

Economic: some farmers lack capital for land preparation; cultural: some farmers have not embraced agriculture as a business and do not use high quality inputs

Economic: some farmers lack funds to purchase insecticides or for transportation; **biophysical:** some farmers live in areas with poor terrain, mainly in Tindiret and some parts of Masop (Cheptawai)

Institutional: farmers lack information on alternative markets; **biophysical:** farmers living in hilly terrains are not able to access good markets

Chicken (Local)



Provision of
Inputs



On-Farm
Production



Harvesting
Storage and
Processing



Product
Marketing



Excess Rainfall
Consequences

High costs production due to expensive construction materials; high chick mortalities during transportation; impassable roads increase transportation costs for inputs; spoilage and loss of feeds during transportation

High brooding costs; intensive management increases labor costs; poor roads increase cost and decrease quality of inputs; high feed consumption; low egg production; impassable roads delay vaccines delivery; high incidences of diseases

Egg production reduced; poor roads cause egg breakage and loss in transport; increased costs for packaging materials; reduced egg quality; reduced prices; increased costs of transportation due to impassable roads

Lack of access to public places like markets delays information transfer; middlemen take advantage of the small-scale farmers who cannot access markets easily; poor roads render marketplaces inaccessible; reduced delivery of eggs

Magnitude of
Impact

Moderate-Severe

Moderate-Major

Minor-Moderate

Minor-Moderate

Farmers' Current
Coping Strategies

Building structures from locally available materials (e.g., off-cuts, old mabatis, grass-thatched houses); maintenance of feeder roads by county government; hatching farm-produced eggs; feed formulation with maize, vegetables, and kitchen leftovers; use of small scale feed millers

Using improved brooders; selection of good brooding hens; use of small feed meals and kitchen leftover or vegetation; farmers treat flocks with herbs

Use of baskets and carton boxes for storage and transport; farm-gate selling

Sourcing information from fellow farmers and extension officer by word of mouth or phone; negotiating directly with institutions (e.g. hotels); hawking eggs and birds; selling directly to chicken vendors; farm-gate selling

Potential
Adaption Options

Upscale and building capacity for the use of appropriate structures; improved road maintenance; hatching chicks at the farm or cooperative level; promotion of on-farm feed formulation

Upscaling the use of brooding technologies (e.g., commercial brooders, biogas and solar-powered brooders); promotion of synchronized and serialized hatching; promoting on-farm feed formulation; building the capacity of farmers regarding diseases and pest control; acquisition of vaccines in bulk; storing vaccines near farmers

Promoting use of appropriate location and construction of laying nests; promoting use of recommended egg trays; promoting aggregation of eggs and birds; promoting value addition in the value chain

Promotion of market information platforms; promoting the use of aggregation; promoting organized chicken and chicken product marketing; promoting bulk selling

Underlying Factors

Infrastructure: most small-scale poultry farmers are a road network rendered impassable during the rains; increased transportation costs; **economic:** small-scale farmers lack capital for poultry houses; **social/cultural:** in most households, poultry is a women-oriented enterprise to which few resources are directed; **policy:** the quality and pricing of feeds

Social/cultural: low literacy levels correlate with inadequate brooding, vaccination, and on-farm feed production knowledge; **economic:** many small-scale farmers lack reliable heat sources

Infrastructure: poor road networks hinder safe transportation; lack of proper chicken markets, abattoirs, and egg aggregation centers hinders value addition; volumes are insufficient to attract large buyers or contract farming

Institutional: lack of organized farmer groups leads to exploitation by middlemen or low farm-gate prices; lack of collection or aggregation centers hinders price negotiation; **policy:** there is no policy in the county detailing how the production, handling, and selling of indigenous chicken and product should be managed, thus the value chain is haphazard



Very Low Temperatures
Consequences

Increased cost of inputs; construction of shelters is labor- and capital-intensive; high mortality rates; increased consumption of feed

High costs of brooding; high consumption of feeds; increased amount of feeds is costly to farmers; high incidences of respiratory diseases

Reduced production due to low temperatures; reduced quantity of saleable eggs; job losses among transporters

Reduced travel reduces information access; low quantities reduce prices; reduced supply

Magnitude of
Impact

Moderate-Severe

Moderate-Severe

Minor-Severe

Minor-Moderate

Farmers' Current
Coping Strategies

Constructing poultry houses with locally available materials; farmers hatching own chicks instead of buying them; synchronized hatching; use of free range systems; feed formulation with locally available food (e.g., crushed maize, kitchen leftovers)

Using improved brooders; using good mother hens; free range systems; feed formulation with scraps; using local herbs to treat diseases

Among a small number of farmers, insulating poultry houses; using plastic egg trays for packaging; sorting eggs according to size and shape; value addition and hawking

Using cell phones for information gathering and price negotiation; value addition

Potential
Adaption Options

Scaling up, and building capacity for appropriate chicken housing; promoting local chick production using incubators; Promoting on-farm feed production; promoting the use of alternative feeds

Promoting the use of green energy for brooding; Promoting feed formulation at farm level; using alternative feed sources; supporting the acquisition of feeds at cooperative level; promoting bulk acquisition of vaccines; building the capacity of farmers on vaccination

Promoting the use of carbohydrate feed sources (e.g., maize); promoting the use of proper housing structures; scaling up the use of plastic egg trays; promoting the grading of eggs for different prices; promoting aggregating and bulking of eggs and chicken for transport

Promoting the use of marketing information platforms; promoting the aggregation and bulking of eggs and chicken; encouraging the formation of producer/marketing organizations;

Underlying Factors

Economic: most small-scale farmers cannot afford to build good poultry houses; **social/cultural:** poultry keeping is mostly practiced by women and youth, who lack production assets and resources; **institutional:** lack of financial partners and development organizations to support poultry farmers through grants

Institutional: most farmers lack brooding knowledge and skills; insufficient corps of frontline extension workers; vaccines are acquired over distances

Social/cultural: women practice poultry but cannot access extension advisory services

Economic: most farmers are not able to brand their products or source market information due to economic constraints; **policy:** no platform to negotiate for better prices or source of market information



Figure 12: Climate variabilities and adaptation strategies across selected value chains in Nandi County

5. Policies and Strategies in climate change

Nandi County bases most of its climate change policies on Kenya's national policies. The Climate Change Act of 2016 and the National Adaptation Plan of Kenya, 2015-2030 require county governments to integrate climate change actions, interventions, and duties into their development plans (Table 1).

Table 1: National policies and strategies targeting climate change adaptation and mitigation

Policy	Year	Policy Objective(s)	Interventions	Challenges
National Climate Change Action Plan	2013-2017	Drafting a climate change policy that includes a proposal to form a climate change department Increasing forest cover and rehabilitating degraded lands	Reclaiming riparian land Formulating by-laws to control deforestation Planting trees Raising awareness among communities	- Limited budget - Inadequate capacity building on climate change issues - Lack of political goodwill - Weak coordination between the public and private sectors - Poor coordination - Weak links in enforcement of regulations such as forest encroachment
National Climate Finance Policy	2016	Establishing mechanisms to mobilize climate change-related finance	Drafting the County Climate Change Fund	
The Kenya Forest Conservation and Management Act	2016	Establishing the Kenya Forestry Service, which protects and manages water catchment areas	Establishing the Forest Conservation and Management Trust Fund Providing education and training on forestry	
Climate Change Act	2016	Guiding Kenya's climate change response by mainstreaming climate change Increasing food security through sustainable production systems	Acting as the legal foundation of the National Climate Change Action Plan Enhancing water security for climate change resilience and strengthening the adaptive capacity of vulnerable groups	
National Climate Change Action Plan	2018-2022	Studying the impacts of climate change on socio-economic sectors and identifying strategic areas where climate action is linked to the "Big Four" agenda pillars of food and nutrition security, manufacturing, universal health care, and affordable housing	Establishing afforestation and reafforestation programs Initiating a partnership with the Green Belt Movement that takes a holistic approach to development by focusing on environmental conservation, community development, and capacity building for the county's climate change action plan	
Kenya Climate Smart Agriculture Strategy	2017-2026	Adapting to climate change and building resilient agricultural systems	Establishing good agricultural practises that lead to enhanced food and nutritional security and improved livelihoods	
The King'wal Integrated Wetland Management Plan	2014-2018	Using wetland resources effectively and efficiently	Halting the current degradation of the wetlands, regulating water flow, groundwater recharge, water storage, and nutrient and pollutant filtering, and stabilizing shorelines and microclimates	

Policy	Year	Policy Objective(s)	Interventions	Challenges
Agriculture Sector Development Strategy, ASDS	2010-2020	Transforming crop, livestock, and fishery production into commercially-oriented enterprises that ensure sustainable food and nutrition security	Supporting the dairy cattle, fish farming, indigenous chicken, and maize value chains	Lack of a good county policy that aligns to national policies
Kenya Cereals Enhancement Program (KCEP)	2014	Supporting farmers to increase the productivity and profitability of maize	Building warehouses in Mosop and Chesumei sub-counties to store maize for farmers Managing the installation of solar dryers to prevent aflatoxin contamination and post-harvest losses Helping farmers market their maize	Slow policy implementation

6. Institutional Capacity on Climate Change

Climate change threatens the agricultural sector of Nandi County. Both government and non-governmental institutions are responsible for managing climate change. In Nandi County, the Department of Agriculture, Livestock, Fisheries and Cooperatives and the Department of Lands, Environment, and Natural Resources support climate change action. They oversee policy formulation, the provision of extension services, and policy enforcement and oversight of climate change management. These departments are supported by other actors, including the Kenya Forestry Service and the National Environmental Management Agency (Table 2).

Table 2: Institutions that are currently supporting and implementing agricultural interventions in Nandi County

Off-Farm Services	Institutions	Interventions in Nandi County	Challenges
Agricultural research and aid services	Department of Agriculture and Cooperative Development	Providing on-farm training and demonstrations on new technologies, innovations, and management practices	Inadequate budget allocations towards climate change by the county government
	Department of Land, Environment, Natural Resources, and Sanitation	Drafting climate change policy bill Implementing the Nandi County Water Provision Master Plan Rehabilitating community water schemes Connecting households to piped water Drilling boreholes Constructing additional water-harvesting structures like tanks	Climate adaptation planning and management is reactive rather than proactive
	National Environment Management Authority	Promoting sustainable environmental management by integrating environmental considerations into policies, plans, programs and projects	Poor coordination among the organizations that leads to duplication and overlap of activities geared towards climate change
	Kenya Forestry Service	Promoting conservation and protection of water-catchment areas, including Nandi's 6 gazetted forests Educating the public on forest conservation Launching a tree-planting program under the National Tree Planting Campaign Formulating policies aimed at protecting carbon sinks Giving forest-adjacent communities portions of land on which to plant trees through a program called the Plantation Establishment and Livelihood Improvement Scheme	Lack of a climate change department that streamlines the activities of all actors.
	Food and Agriculture Organization	Assisting the county government in developing the Nandi County Wetland Management Plan, which focuses on the challenges facing wetlands across the county	

Off-Farm Services	Institutions	Interventions in Nandi County	Challenges
	Kenya Tea Development Authority	Managing smallholder tea farmers in Nandi County Providing subsidised inputs, establishing farmer field schools, Coordinating post-production activities, marketing, and payments	Deliberate operational planning for climate change is still limited
	Multinational tea corporation (Eastern Produce Kenya, Williamson Tea)	Producing and processing tea in large quantities	Inadequate personnel in the extension service across the county.
	Anglican Development Services	Supporting the Department of Agriculture in educating farmers	
	International Fund for Agricultural Development	Implementing the Smallholder Dairy Commercialization Program, which strengthens dairy farmers' organizations	The central government dominates issues of Policy formulation and other actors are inadequately involved
	Directorate of Livestock and Private Veterinary Officers	Offering AI services Selecting animals that suit different agroecological zones	
Climate Information Services and Weather Advisories	Kenya Meteorological Department	Issuing weekly bulletins and seasonal weather forecasts in the local language, via local radio and TV stations, social media platforms like WhatsApp, and SMS messages	Farmers' cooperative societies and associations are often weak or disorganized with no clear governance structure, leaving farmers susceptible to exploitation by intermediaries
Early Warning Systems and Participatory Scenario Planning	Kenya Meteorological Department	Issuing warnings on disasters like mudslides and landslides Issuing advisories and weather updates Supporting farm-activity planning	
Non-financial Subsidies	National Cereals and Produce Board	Providing subsidized farm inputs like certified seeds Providing maize markets	
	Unga feeds, Cooper K Brand, East Africa Seeds Co. Ltd., Kenya Seeds Co., Osho Chemicals, Bayer East Africa, Baraka Fertilizers Ltd., Agrovets	Distributing and selling agro-chemicals and other farm inputs, as well as providing training and demonstration on the use of these inputs	Inadequate loan facilities available to smallholder farmers.
Financial Services	Savings and Credit Cooperative Societies	Providing farmers with savings and credit facilities	
	Banks	Providing farmers with a platforms for savings, loans, and transactions	
	Microfinance Societies	Providing short-term and long-term loans to farmers	
	Insurance Societies	Acre Africa has partnered with the Nandi County Government to provide farmers with weather-index insurance coverage.	
Market Services	New Kenya Cooperative Creameries Brookside Dairy Ltd.	Boosting milk sales and processing Processing and marketing milk	

7. Synthesis and Outlook

Although agriculture is the economic mainstay of Nandi County, its full potential has not been realized. Maize, tea, dairy cattle, and indigenous chickens are all affected by climate variability that has led to changes in planting seasons, moisture stress, excess rainfall, hailstorms, frost, and very low temperatures. Farmers are adapting to climate change, but their options are mainly located on-farm. Identifying and growing new technologies could harness the adaptive capacities of different actors.

Nandi County's reliance on firewood is an urgent threat to forests, which act as water-catchment areas and carbon sinks. Policies might be established and enforced to protect these areas. All government departments, for example, could emphasize the national policy of 10% forest cover. The promotion and adoption of modern technologies like drip irrigation systems as well as livestock feed conservation technologies including silage making and hay production can help save the county's forests.

Although government initiatives are partnering with the private sector to provide weather-index insurance for the farmers of Nandi County, uptake is still very low. Farmers could be sensitized to the benefits of insurance and helped to procure it. Because county governments primarily support farmers through

cooperatives, it is vital to encourage all farmers to join. Moreso, multi-purpose cooperatives for value chains that complement each other need to be established. Extension services will go a long way towards equipping farmers with knowledge on good agronomic practices and help them minimize post-production losses. Climate advisory services can inform farmers on optimum planting windows and how to avoid crop loss. Promoting early-maturing and drought-tolerant crop varieties and livestock breeds will also foster adaptation.

Additional provision of subsidies is necessary, especially for the acquisition of inputs and technologies to enhance resilience. For example, while chicken farmers in Nandi County hatch their chicks using the conventional method of hens sitting of their eggs, subsidies to facilitate the adoption of incubators will boost hatching rates.

Strong, supportive policy is key for managing climate change. Nandi County's climate change policy awaits approval from the county assembly. A county climate change department could streamline climate change activities, coordinate stakeholder engagement, and enforce current policy to boost resilience to climate change effects.

8. Works Cited

ASDSP. 2014. Nandi County.

Bore, J. K. 2015. Tea Adaptation and Mitigation to Climate Change in Kenya. Tea Research Institute, Kericho, Kenya: Kenya Livestock and Research Organization (KALRO).

County Government of Nandi. 2013. Nandi County Integrated Development Plan (CIDP) 2013-2017, Nandi, Kenya

County Government of Nandi. 2018. Nandi County Integrated Development Plan (CIDP) 2018-2022. Nandi, Kenya

County Government of Nandi. 2020. FIELD DAY-Enhancing Food Security and Income through Partnership.- <https://bit.ly/3ftGJ89>

Gachenge, B. 2012. Frost Curbs Kenya's January Tea Production. Reuters

Retrieved from(2012)- <https://reut.rs/3urhCH9>

IPCC, 2018: Global warming of 1.5°C. An IPCC Special Report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change, sustainable development, and efforts to eradicate poverty [V. Masson-Delmotte, P. Zhai, H. O. Pörtner, D. Roberts, J. Skea, P.R. Shukla, A. Pirani, W. Moufouma-Okia, C. Péan, R. Pidcock, S. Connors, J. B. R. Matthews, Y. Chen, X. Zhou, M. I. Gomis, E. Lonnoy, T. Maycock, M. Tignor, T. Waterfield (eds.)]. In Press.

KCEP. 2016. Adaptation and dissemination of available technologies for smallholder adoption, KCEP-KALRO maize production training and extension manual. Nairobi, Kenya

KMD. (2020). State of the Climate in Kenya 2020.

KNBS. 2019a. Distribution of the population by Socio-Economic Characteristics, Nairobi, Kenya

KNBS. 2019b. Kenya Population and Housing Census, Nairobi, Kenya

Navarro-Racines C, Tarapues J, Thornton P, Jarvis A, Ramirez-Villegas J (2020) High-resolution and bias-corrected CMIP5 projections for climate change impact assessments. *Sci Data* 7(1):1–14. <https://bit.ly/3bZJYBU>

Owuor, C. (2020). Ten counties with surplus maize stock. Nairobi, Kenya. Derived from People Daily <https://bit.ly/2Rlf8Hn>

Taylor, K.E., Stouffer, R.J., & Meehl, G.A. (2012). An overview of CMIP5 and the experiment design. *Bulletin of the American Meteorological Society*, 93: 485-498.

Wambugu C, Franzel S, Rioux J 2014. Options for climate-smart agriculture at Kaptumo sit in Kenya. ICRAF Working Paper No. 185. Nairobi, World Agroforestry Centre. DOI: <http://apps.worldagroforestry.org/downloads/Publications/PDFS/WP14394.pdf>

Willson, K. C. (1992). Field Operations: 1. In K. C. Willson (Ed.), *Tea: cultivation to consumption*. London: Chapman & Hall.

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10.1 Glossary

Climate variability: Variations in the mean state and other statistics (such as standard deviations, the occurrence of extremes, etc.) of the climate on all spatial and temporal scales beyond that of individual weather events (IPCC, 2018).

Weather-index insurance- An insurance scheme that pays out based on an index, such as rainfall, measured at a local weather station or by satellite, rather than based on a consequence of weather, such as a farmer's crop yield. Unlike traditional crop insurance, the insurance company does not need to visit a farmer's field to determine premiums or assess damages; if the rainfall amount is below a pre-specified threshold, then the insurance pays out.

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